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Chapter 16 - The Matrix Exponential

Chapter 15 introduced matrix multiplication, powers, eigenvalues, and the Cayley-Hamilton theorem. This chapter combines those ideas to define and compute the matrix exponential:

$$e^{At}.$$

The notation resembles an ordinary exponential, but the exponent contains a matrix. That makes factor order and matrix powers essential.

The main questions are:

- what does e^{At} mean?
- why does its defining series converge?
- why is $\frac{d}{dt}e^{At} = Ae^{At}$?
- how does e^{At} solve $\mathbf{x}' = A\mathbf{x}$?
- when does the infinite series stop after finitely many terms?
- how does Cayley-Hamilton reduce the exponential to a matrix polynomial?
- what changes when eigenvalues repeat?
- how do complex eigenvalues produce real sine and cosine entries?
- which familiar exponential laws remain valid for matrices?

Defining An Exponential Of A Matrix

The Goal Of This Block

The goal is to give e^{At} a precise meaning and to connect its notation to the familiar scalar exponential series.

Recall The Scalar Series

For a scalar z :

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots.$$

Set $z = at$:

$$e^{at} = 1 + at + \frac{a^2 t^2}{2!} + \frac{a^3 t^3}{3!} + \cdots.$$

The matrix exponential uses the same coefficient pattern, but scalar powers a^k are replaced by matrix powers A^k .

The Definition

Let A be a constant $n \times n$ matrix. Define:

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}.$$

Writing the first terms:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots.$$

The term for $k = 0$ is:

$$\frac{A^0 t^0}{0!} = I,$$

because:

$$A^0 = I, \quad t^0 = 1, \quad 0! = 1.$$

Why The Matrix Must Be Square

The series contains:

$$A^2, \quad A^3, \quad A^4, \quad \dots$$

These ordinary matrix powers require the product AA to be defined and to have the same dimensions as A . That happens when A is square.

Every term in the series is then $n \times n$, so the terms can be added.

Convergence Without Hidden Mystery

Choose any standard matrix norm $\|\cdot\|$ satisfying:

$$\|AB\| \leq \|A\| \|B\|.$$

Then:

$$\left\| \frac{A^k t^k}{k!} \right\| \leq \frac{\|A\|^k |t|^k}{k!}.$$

The scalar comparison series is:

$$\sum_{k=0}^{\infty} \frac{(\|A\|t)^k}{k!} = e^{\|A\|t},$$

which converges for every finite t .

Therefore the matrix series also converges for every finite t .

The key intuition:

the factorial in the denominator grows quickly enough to control the matrix powers, just as it controls powers in the ordinary exponential series.

Two Immediate Values

Set $t = 0$ in the series:

$$\begin{aligned} e^{A0} &= I + A(0) + \frac{A^2(0)^2}{2!} + \dots \\ &= I. \end{aligned}$$

Thus:

$$\boxed{e^{A0} = I}.$$

If $A = 0$, then every positive power of A is zero:

$$0^k = 0 \quad \text{for } k \geq 1.$$

Because every positive-power term vanishes, the zero-matrix exponential is:

$$\boxed{e^{0t} = I}.$$

The matrix identity I , not the scalar 1, is the constant term.

Not An Entrywise Exponential

For:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the notation e^{At} does not generally mean:

$$\begin{bmatrix} e^{at} & e^{bt} \\ e^{ct} & e^{dt} \end{bmatrix}.$$

The correct definition uses the matrix products $A^2, A^3, \dots$. Those products mix rows and columns, so one entry of e^{At} can depend on several entries of A .

Entrywise exponentiation is a different operation.

Block 1 Summary

For a constant square matrix A :

$$\boxed{e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots}.$$

The series converges for every finite t , begins at I , and uses matrix powers rather than entrywise powers.

Why The Matrix Exponential Solves A System

The Goal Of This Block

The goal is to prove the derivative identity for e^{At} and use it to solve a constant-coefficient matrix differential equation.

Differentiate The Series Term By Term

Start from:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

Differentiate with respect to t . The matrix A is constant:

$$\begin{aligned} \frac{d}{dt}e^{At} &= 0 + A + \frac{2A^2 t}{2!} + \frac{3A^3 t^2}{3!} + \dots \\ &= A + A^2 t + \frac{A^3 t^2}{2!} + \dots \end{aligned}$$

Factor A on the left:

$$\begin{aligned} \frac{d}{dt}e^{At} &= A \left(I + At + \frac{A^2 t^2}{2!} + \dots \right) \\ &= Ae^{At}. \end{aligned}$$

Therefore:

$$\boxed{\frac{d}{dt}e^{At} = Ae^{At}}.$$

Why The Matrix Can Also Appear On The Right

Every term in e^{At} is a scalar multiple of a power of A . Powers of the same matrix commute:

$$AA^k = A^{k+1} = A^k A.$$

Therefore:

$$Ae^{At} = e^{At}A.$$

The derivative may also be written:

$$\boxed{\frac{d}{dt}e^{At} = e^{At}A}.$$

This does not mean that A commutes with every matrix. It commutes here because e^{At} is built entirely from powers of A .

Solving The Initial-Value Problem

Consider:

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad \mathbf{x}(t_0) = \mathbf{c}.$$

Propose:

$$\boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{c}}.$$

For $\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{c}$, differentiate using the matrix exponential derivative:

$$\begin{aligned}\mathbf{x}'(t) &= \frac{d}{dt} [e^{A(t-t_0)}] \mathbf{c} \\ &= A e^{A(t-t_0)} \mathbf{c} \\ &= A \mathbf{x}(t).\end{aligned}$$

Now check the initial value. Set $t = t_0$:

$$\begin{aligned}\mathbf{x}(t_0) &= e^{A(t_0-t_0)} \mathbf{c} \\ &= e^{A0} \mathbf{c} \\ &= I \mathbf{c} \\ &= \mathbf{c}.\end{aligned}$$

The proposed formula satisfies both the differential equation and the initial condition.

The Columns Are Individual Solutions

Write:

$$e^{At} = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1(t) & \mathbf{x}_2(t) & \cdots & \mathbf{x}_n(t) \\ | & | & & | \end{bmatrix}.$$

Since:

$$\frac{d}{dt} e^{At} = A e^{At},$$

Because $(e^{At})' = Ae^{At}$, equality of corresponding columns gives:

$$\mathbf{x}'_j(t) = A\mathbf{x}_j(t)$$

for each j .

At $t = 0$:

$$e^{A0} = I.$$

Therefore the j th column starts from the j th standard basis vector.

What this tells us:

one matrix e^{At} packages a complete set of specially normalized solutions of the system.

A Diagonal System Example

Let:

$$D = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$$

and solve:

$$\mathbf{x}' = D\mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

For $D = \text{diag}(2, -3)$, the powers remain diagonal:

$$D^k = \begin{bmatrix} 2^k & 0 \\ 0 & (-3)^k \end{bmatrix}.$$

Substitute these powers into the defining series:

$$\begin{aligned} e^{Dt} &= \begin{bmatrix} \sum_{k=0}^{\infty} \frac{2^k t^k}{k!} & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{(-3)^k t^k}{k!} \end{bmatrix} \\ &= \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{-3t} \end{bmatrix}. \end{aligned}$$

Use $\mathbf{x}(t) = e^{Dt}\mathbf{x}(0)$:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{-3t} \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 4e^{2t} \\ -e^{-3t} \end{bmatrix}. \end{aligned}$$

The solution is:

$$\boxed{\mathbf{x}(t) = \begin{bmatrix} 4e^{2t} \\ -e^{-3t} \end{bmatrix}}.$$

The first component grows, while the second decays.

Block 2 Summary

The matrix exponential satisfies:

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$$

and:

$$e^{A0} = I.$$

Therefore:

$$\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{c}$$

solves $\mathbf{x}' = A\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{c}$.

Diagonal And Nilpotent Shortcuts

The Goal Of This Block

The goal is to recognize matrices for which the exponential can be read from the series without solving an interpolation problem.

Diagonal Matrices

For:

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix},$$

For $D = \text{diag}(d_1, \dots, d_n)$, every power is:

$$D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix}.$$

Substituting into the exponential series gives:

$$e^{Dt} = \begin{bmatrix} e^{d_1 t} & 0 & \cdots & 0 \\ 0 & e^{d_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{d_n t} \end{bmatrix}.$$

Only diagonal matrices permit this simple entrywise-looking rule.

Nilpotent Matrices

A matrix N is **nilpotent** if some positive power equals zero.

The defining condition is:

$$N^r = 0,$$

Once $N^r = 0$, every higher power also vanishes:

$$N^{r+1} = N^r N = 0, \quad N^{r+2} = 0, \quad \dots$$

The exponential series stops:

$$e^{Nt} = I + Nt + \frac{N^2 t^2}{2!} + \cdots + \frac{N^{r-1} t^{r-1}}{(r-1)!}.$$

The infinite series has become a finite matrix polynomial.

A Nilpotent Example

Let:

$$N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Calculate N^2 :

$$\begin{aligned} N^2 &= \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0(0) + 2(0) & 0(2) + 2(0) \\ 0(0) + 0(0) & 0(2) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Since $N^2 = 0$, every term from the quadratic term onward vanishes:

$$\begin{aligned} e^{Nt} &= I + Nt + \frac{N^2 t^2}{2!} + \dots \\ &= I + Nt. \end{aligned}$$

Substitute I and N :

$$\begin{aligned} e^{Nt} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + t \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

The nilpotent exponential is:

$$e^{Nt} = \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

A Scalar Shift Of A Nilpotent Matrix

Consider:

$$A = \begin{bmatrix} 4 & 2 \\ 0 & 4 \end{bmatrix}.$$

Separate the scalar part:

$$A = 4I + N,$$

where:

$$N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

and $N^2 = 0$.

The matrices $4I$ and N commute. For commuting matrices, the exponential of a sum splits into a product:

$$e^{(4I+N)t} = e^{4It}e^{Nt}.$$

This exponential law will be justified in Block 7.

For the commuting split $A = 4I + N$, the two exponential factors are:

$$e^{4It} = e^{4t}I$$

and:

$$e^{Nt} = \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix},$$

we obtain:

$$e^{At} = e^{4t} \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

Verify The Shifted Formula

Let:

$$E(t) = e^{4t} \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

Differentiate using the scalar product rule:

$$\begin{aligned} E'(t) &= 4e^{4t} \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} + e^{4t} \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \\ &= e^{4t} \begin{bmatrix} 4 & 8t + 2 \\ 0 & 4 \end{bmatrix}. \end{aligned}$$

Recall the candidate:

$$E(t) = e^{4t} \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

Now calculate $AE(t)$:

$$\begin{aligned} AE(t) &= \begin{bmatrix} 4 & 2 \\ 0 & 4 \end{bmatrix} e^{4t} \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} \\ &= e^{4t} \begin{bmatrix} 4 & 8t + 2 \\ 0 & 4 \end{bmatrix}. \end{aligned}$$

The derivative and matrix product agree:

$$E'(t) = AE(t).$$

At $t = 0$:

$$E(0) = I.$$

These two checks confirm that $E(t) = e^{At}$.

Block 3 Summary

For a diagonal matrix, exponentiate the diagonal entries. For a nilpotent matrix N with $N^r = 0$, the defining series stops after the term containing N^{r-1} .

When a matrix has the scalar-plus-nilpotent form:

$$A = \alpha I + N,$$

Because αI commutes with N , a matrix $A = \alpha I + N$ satisfies:

$$e^{At} = e^{\alpha t} e^{Nt},$$

This turns the matrix calculation into a scalar exponential multiplied by a finite nilpotent exponential.

Cayley-Hamilton With Distinct Eigenvalues

The Goal Of This Block

The goal is to replace the infinite series for e^{At} with a finite polynomial in A and calculate its coefficients from distinct eigenvalues.

Why A Finite Polynomial Is Possible

Let A be $n \times n$, with characteristic polynomial:

$$p_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0.$$

Cayley-Hamilton gives:

$$p_A(A) = 0.$$

Therefore A^n can be written using lower powers:

$$A^n = -c_{n-1}A^{n-1} - \dots - c_0I.$$

Multiplying this identity by further powers of A reduces $A^{n+1}, A^{n+2}, \dots$ as well.

Consequently, the matrix exponential can be written in the form:

$$e^{At} = a_0(t)I + a_1(t)A + \cdots + a_{n-1}(t)A^{n-1}.$$

The functions $a_0, \dots, a_{n-1}$ depend on t and on the matrix A .

The Interpolating Polynomial

Introduce the scalar polynomial:

$$r_t(\lambda) = a_0(t) + a_1(t)\lambda + \cdots + a_{n-1}(t)\lambda^{n-1}.$$

The subscript t reminds us that t is held fixed while λ is the polynomial variable.

For every distinct eigenvalue λ_j of A , require:

$$r_t(\lambda_j) = e^{\lambda_j t}.$$

Once the scalar coefficient functions have been found, replace λ by A :

$$e^{At} = r_t(A).$$

This separates the calculation into two stages:

1. solve a scalar interpolation problem;
2. evaluate the resulting polynomial at the matrix.

A Complete Distinct-Eigenvalue Example

Let:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

We will find e^{At} without finding eigenvectors.

For the matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix},$$

construct:

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & -1 \\ -3 & \lambda - 4 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= (\lambda - 2)(\lambda - 4) - (-1)(-3) \\ &= (\lambda - 2)(\lambda - 4) - 3. \end{aligned}$$

Expand:

$$\begin{aligned} p_A(\lambda) &= \lambda^2 - 6\lambda + 8 - 3 \\ &= \lambda^2 - 6\lambda + 5. \end{aligned}$$

Factor:

$$p_A(\lambda) = (\lambda - 1)(\lambda - 5).$$

Therefore the two distinct eigenvalues are:

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 5}.$$

Build The Coefficient Equations

Because A is 2×2 , use a polynomial of degree at most 1:

$$r_t(\lambda) = a_0 + a_1\lambda.$$

The coefficient functions depend on t , but the shorter notation a_0, a_1 will be used inside this calculation.

For the distinct eigenvalues 1 and 5, begin at $\lambda_1 = 1$ and impose:

$$r_t(1) = e^t.$$

Substitute $\lambda = 1$ into $a_0 + a_1\lambda$:

$$a_0 + a_1 = e^t.$$

At $\lambda_2 = 5$, impose:

$$r_t(5) = e^{5t}.$$

Substitute $\lambda = 5$:

$$a_0 + 5a_1 = e^{5t}.$$

Thus the coefficient system is:

$$\begin{aligned}a_0 + a_1 &= e^t, \\a_0 + 5a_1 &= e^{5t}.\end{aligned}$$

Solve The Coefficient System

Subtract the first equation from the second:

$$(a_0 + 5a_1) - (a_0 + a_1) = e^{5t} - e^t.$$

The a_0 terms cancel:

$$4a_1 = e^{5t} - e^t.$$

From $4a_1 = e^{5t} - e^t$, divide by 4:

$$a_1 = \frac{e^{5t} - e^t}{4}.$$

Return to:

$$a_0 + a_1 = e^t.$$

Subtract a_1 :

$$a_0 = e^t - a_1.$$

Insert the calculated a_1 :

$$\begin{aligned} a_0 &= e^t - \frac{e^{5t} - e^t}{4} \\ &= \frac{4e^t - e^{5t} + e^t}{4} \\ &= \frac{5e^t - e^{5t}}{4}. \end{aligned}$$

Therefore:

$$a_0 = \frac{5e^t - e^{5t}}{4}.$$

Evaluate The Polynomial At The Matrix

The reduced exponential is:

$$e^{At} = a_0 I + a_1 A.$$

Substitute a_0 , a_1 , I , and A :

$$\begin{aligned} e^{At} &= \frac{5e^t - e^{5t}}{4} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &\quad + \frac{e^{5t} - e^t}{4} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}. \end{aligned}$$

Calculate each entry. The row 1, column 1 entry is:

$$\begin{aligned} a_0 + 2a_1 &= \frac{5e^t - e^{5t}}{4} + \frac{2e^{5t} - 2e^t}{4} \\ &= \frac{3e^t + e^{5t}}{4}. \end{aligned}$$

The row 1, column 2 entry is:

$$a_1 = \frac{e^{5t} - e^t}{4}.$$

The row 2, column 1 entry is:

$$3a_1 = \frac{3e^{5t} - 3e^t}{4}.$$

The row 2, column 2 entry is:

$$\begin{aligned} a_0 + 4a_1 &= \frac{5e^t - e^{5t}}{4} + \frac{4e^{5t} - 4e^t}{4} \\ &= \frac{e^t + 3e^{5t}}{4}. \end{aligned}$$

Place the entries into the matrix:

$$e^{At} = \frac{1}{4} \begin{bmatrix} 3e^t + e^{5t} & e^{5t} - e^t \\ 3e^{5t} - 3e^t & e^t + 3e^{5t} \end{bmatrix}.$$

Use The Result In An Initial-Value Problem

Solve:

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

Use the matrix-exponential solution formula:

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0).$$

Substitute the calculated exponential:

$$\begin{aligned}\mathbf{x}(t) &= \frac{1}{4} \begin{bmatrix} 3e^t + e^{5t} & e^{5t} - e^t \\ 3e^{5t} - 3e^t & e^t + 3e^{5t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 3e^t + e^{5t} \\ 3e^{5t} - 3e^t \end{bmatrix}.\end{aligned}$$

The solution selected by the initial vector is:

$$\mathbf{x}(t) = \frac{1}{4} \begin{bmatrix} 3e^t + e^{5t} \\ 3e^{5t} - 3e^t \end{bmatrix}.$$

Setting $t = 0$ gives:

$$\mathbf{x}(0) = \frac{1}{4} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so the initial condition is satisfied.

Block 4 Summary

Cayley-Hamilton reduces e^{At} to a polynomial of degree at most $n - 1$ in an $n \times n$ matrix.

For distinct eigenvalues, build:

$$r_t(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1}$$

For each distinct eigenvalue λ_j , impose:

$$r_t(\lambda_j) = e^{\lambda_j t}.$$

Solve for the scalar coefficient functions, then calculate:

$$e^{At} = r_t(A).$$

Repeated Eigenvalues And Derivative Conditions

The Goal Of This Block

The goal is to explain why a repeated eigenvalue supplies more than one coefficient equation and to keep the differentiation variable explicit.

Why Repeating The Same Value Is Not Enough

Suppose a 2×2 matrix has only one eigenvalue λ_0 , repeated twice. The reduced polynomial is:

$$r_t(\lambda) = a_0 + a_1 \lambda.$$

The value condition:

$$r_t(\lambda_0) = e^{\lambda_0 t}$$

provides only one equation for two unknown coefficient functions.

Writing that same equation twice adds no information. A second, independent condition is needed.

The Derivative-Matching Rule

If λ_0 is an eigenvalue of algebraic multiplicity m , match the first m pieces of derivative data:

$$\boxed{\frac{\partial^q r_t}{\partial \lambda^q}(\lambda_0) = \frac{\partial^q}{\partial \lambda^q} e^{\lambda t} \Big|_{\lambda=\lambda_0}, \quad q = 0, 1, \dots, m-1.}$$

The differentiation is with respect to λ . During this operation, t is treated as a constant.

For the exponential:

$$\frac{\partial}{\partial \lambda} e^{\lambda t} = t e^{\lambda t},$$

Differentiating once more gives:

$$\frac{\partial^2}{\partial \lambda^2} e^{\lambda t} = t^2 e^{\lambda t}.$$

A repeated eigenvalue of multiplicity 2 requires value and first-derivative matching. Multiplicity 3 requires value, first derivative, and second derivative.

A Complete Repeated-Eigenvalue Example

Let:

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}.$$

Construct:

$$\lambda I - A = \begin{bmatrix} \lambda - 3 & -1 \\ 1 & \lambda - 1 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= (\lambda - 3)(\lambda - 1) - (-1)(1) \\ &= (\lambda - 3)(\lambda - 1) + 1. \end{aligned}$$

Expanding $p_A(\lambda) = (\lambda - 3)(\lambda - 1) + 1$ gives:

$$\begin{aligned} p_A(\lambda) &= \lambda^2 - 4\lambda + 3 + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2. \end{aligned}$$

Thus $\lambda = 2$ has algebraic multiplicity 2.

Build Both Coefficient Conditions

For a 2×2 matrix, use:

$$r_t(\lambda) = a_0 + a_1\lambda.$$

First match values at $\lambda = 2$:

$$r_t(2) = e^{2t}.$$

Substitute $\lambda = 2$:

$$a_0 + 2a_1 = e^{2t}.$$

Now differentiate with respect to λ :

$$\frac{\partial r_t}{\partial \lambda} = a_1.$$

Differentiate the target function with respect to λ :

$$\frac{\partial}{\partial \lambda} e^{\lambda t} = t e^{\lambda t}.$$

Since $\partial r_t / \partial \lambda = a_1$ and $\partial e^{\lambda t} / \partial \lambda = t e^{\lambda t}$, evaluation at $\lambda = 2$ gives:

$$a_1 = t e^{2t}.$$

The two independent equations are therefore:

$$\begin{aligned} a_0 + 2a_1 &= e^{2t}, \\ a_1 &= t e^{2t}. \end{aligned}$$

Solve And Evaluate At The Matrix

The second equation already gives:

$$a_1 = t e^{2t}.$$

Substitute this into the first equation:

$$a_0 + 2t e^{2t} = e^{2t}.$$

Subtract $2te^{2t}$:

$$\begin{aligned}a_0 &= e^{2t} - 2te^{2t} \\ &= e^{2t}(1 - 2t).\end{aligned}$$

Therefore:

$$e^{At} = a_0I + a_1A.$$

Insert $a_0 = e^{2t}(1 - 2t)$ and $a_1 = te^{2t}$:

$$\begin{aligned}e^{At} &= e^{2t}(1 - 2t)I + te^{2t}A \\ &= e^{2t}[(1 - 2t)I + tA].\end{aligned}$$

Now substitute I and A :

$$\begin{aligned}(1 - 2t)I + tA &= (1 - 2t) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + t \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 - 2t + 3t & t \\ -t & 1 - 2t + t \end{bmatrix} \\ &= \begin{bmatrix} 1 + t & t \\ -t & 1 - t \end{bmatrix}.\end{aligned}$$

Hence:

$$\boxed{e^{At} = e^{2t} \begin{bmatrix} 1 + t & t \\ -t & 1 - t \end{bmatrix}}.$$

Verify The Repeated-Eigenvalue Formula

Let:

$$E(t) = e^{2t} \begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix}.$$

At $t = 0$:

$$E(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Differentiate. Let:

$$M(t) = \begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix}.$$

Then:

$$M'(t) = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Use the product rule:

$$E'(t) = e^{2t} [2M(t) + M'(t)].$$

Calculate the bracket:

$$\begin{aligned} 2M + M' &= 2 \begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3+2t & 1+2t \\ -1-2t & 1-2t \end{bmatrix}. \end{aligned}$$

Now calculate $AM(t)$:

$$\begin{aligned} AM &= \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix} \\ &= \begin{bmatrix} 3+2t & 1+2t \\ -1-2t & 1-2t \end{bmatrix}. \end{aligned}$$

Because the differentiated matrix and $AM(t)$ agree:

$$E'(t) = e^{2t}AM(t) = AE(t).$$

Together with $E(0) = I$, this verifies the exponential.

Do Not Differentiate With Respect To The Wrong Variable

The interpolation polynomial is:

$$r_t(\lambda) = a_0(t) + a_1(t)\lambda.$$

When repeated-eigenvalue conditions are formed, differentiate with respect to λ :

$$\frac{\partial r_t}{\partial \lambda} = a_1(t).$$

Differentiating with respect to t would instead give:

$$\frac{\partial r_t}{\partial t} = a'_0(t) + a'_1(t)\lambda,$$

which is a different expression and does not provide the required Hermite interpolation condition.

What to remember:

hold t fixed, differentiate in λ , and only then substitute the repeated eigenvalue.

Block 5 Summary

For an eigenvalue λ_0 of multiplicity m , match:

$$r_t^{(q)}(\lambda_0) = \left. \frac{\partial^q}{\partial \lambda^q} e^{\lambda t} \right|_{\lambda=\lambda_0}, \quad q = 0, \dots, m-1.$$

These are derivatives with respect to λ , not t . They provide the additional independent equations needed to determine the polynomial coefficients.

Complex Eigenvalues And Real Rotation Matrices

The Goal Of This Block

The goal is to see how a real matrix with complex eigenvalues can have a completely real matrix exponential built from sine and cosine.

A Matrix Whose Square Is Negative

Let:

$$J = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}.$$

Calculate J^2 :

$$\begin{aligned} J^2 &= \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} \\ &= -4I. \end{aligned}$$

The identity $J^2 = -4I$ plays the same role as:

$$i^2 = -1$$

in the scalar exponential.

Separate Even And Odd Powers

Start from:

$$e^{Jt} = I + Jt + \frac{J^2 t^2}{2!} + \frac{J^3 t^3}{3!} + \cdots.$$

Since $J^2 = -4I$:

$$J^{2m} = (-4)^m I$$

and:

$$J^{2m+1} = (-4)^m J.$$

Group the even powers:

$$\begin{aligned} \sum_{m=0}^{\infty} \frac{J^{2m} t^{2m}}{(2m)!} &= I \sum_{m=0}^{\infty} \frac{(-4)^m t^{2m}}{(2m)!} \\ &= I \sum_{m=0}^{\infty} \frac{(-1)^m (2t)^{2m}}{(2m)!} \\ &= I \cos(2t). \end{aligned}$$

The remaining odd-power terms are:

$$\begin{aligned}\sum_{m=0}^{\infty} \frac{J^{2m+1} t^{2m+1}}{(2m+1)!} &= J \sum_{m=0}^{\infty} \frac{(-4)^m t^{2m+1}}{(2m+1)!} \\ &= \frac{J}{2} \sum_{m=0}^{\infty} \frac{(-1)^m (2t)^{2m+1}}{(2m+1)!} \\ &= \frac{J}{2} \sin(2t).\end{aligned}$$

Add the even and odd parts:

$$e^{Jt} = I \cos(2t) + \frac{J}{2} \sin(2t).$$

Write The Real Matrix Explicitly

Substitute:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and:

$$\frac{J}{2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then:

$$\begin{aligned}e^{Jt} &= \begin{bmatrix} \cos(2t) & 0 \\ 0 & \cos(2t) \end{bmatrix} + \begin{bmatrix} 0 & -\sin(2t) \\ \sin(2t) & 0 \end{bmatrix} \\ &= \begin{bmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{bmatrix}.\end{aligned}$$

Combining the even and odd series gives the real matrix:

$$e^{Jt} = \begin{bmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{bmatrix}.$$

This matrix rotates vectors through angle $2t$.

Where The Complex Eigenvalues Appear

Construct:

$$\lambda I - J = \begin{bmatrix} \lambda & 2 \\ -2 & \lambda \end{bmatrix}.$$

Its determinant is:

$$\begin{aligned} p_J(\lambda) &= \lambda^2 - 2(-2) \\ &= \lambda^2 + 4. \end{aligned}$$

Set the polynomial equal to zero:

$$\lambda^2 + 4 = 0.$$

Thus:

$$\lambda = \pm 2i.$$

The eigenvalues are complex, but the sine and cosine terms combine into a real matrix exponential because J itself is real.

Decay Combined With Rotation

Let:

$$A = -I + J = \begin{bmatrix} -1 & -2 \\ 2 & -1 \end{bmatrix}.$$

The matrices $-I$ and J commute, so:

$$e^{At} = e^{-tI}e^{Jt}.$$

Since $e^{-tI} = e^{-t}I$:

$$e^{At} = e^{-t} \begin{bmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{bmatrix}.$$

The factor e^{-t} shrinks lengths, while the sine-cosine matrix rotates.

A Complete Initial-Value Example

Solve:

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 3 \\ 0 \end{bmatrix},$$

where:

$$A = \begin{bmatrix} -1 & -2 \\ 2 & -1 \end{bmatrix}.$$

The initial vector is:

$$\mathbf{x}(0) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Use:

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0).$$

Substitute the exponential:

$$\begin{aligned}\mathbf{x}(t) &= e^{-t} \begin{bmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} \\ &= 3e^{-t} \begin{bmatrix} \cos(2t) \\ \sin(2t) \end{bmatrix}.\end{aligned}$$

Thus:

$$\boxed{\mathbf{x}(t) = 3e^{-t} \begin{bmatrix} \cos(2t) \\ \sin(2t) \end{bmatrix}}.$$

The solution rotates counterclockwise while approaching the origin.

A General Two-By-Two Pattern

Suppose:

$$A = \alpha I + K$$

For this pattern, assume also that:

$$K^2 = -\beta^2 I, \quad \beta > 0.$$

For $A = \alpha I + K$, the matrix αI commutes with K , so:

$$e^{At} = e^{\alpha t} e^{Kt}.$$

The same even-odd series argument gives:

$$e^{At} = e^{\alpha t} \left[I \cos(\beta t) + \frac{K}{\beta} \sin(\beta t) \right].$$

The scalar α controls exponential growth or decay. The scalar β controls the rotation frequency.

Block 6 Summary

A real matrix can have complex eigenvalues and still have a real exponential. When $J^2 = -\beta^2 I$:

$$e^{Jt} = I \cos(\beta t) + \frac{J}{\beta} \sin(\beta t).$$

A scalar shift αI contributes the factor $e^{\alpha t}$.

Exponential Laws, Workflow, And Common Mistakes

The Goal Of This Block

The goal is to identify which scalar exponential laws remain valid, state the required commutativity condition, and organize the available computation methods.

The Same-Matrix Addition Law

The matrices At and As commute because both are scalar multiples of A . Therefore:

$$e^{At}e^{As} = e^{A(t+s)}.$$

The order can also be reversed:

$$e^{As}e^{At} = e^{A(t+s)}.$$

This is the matrix version of the scalar law $e^{at}e^{as} = e^{a(t+s)}$.

The Inverse Of A Matrix Exponential

Set $s = -t$ in the same-matrix addition law:

$$e^{At}e^{-At} = e^{A(t-t)}.$$

Simplify the exponent:

$$e^{At}e^{-At} = e^{A0}.$$

Since $e^{A0} = I$:

$$e^{At}e^{-At} = I.$$

Similarly:

$$e^{-At}e^{At} = I.$$

Therefore e^{At} is always invertible, with:

$$(e^{At})^{-1} = e^{-At}.$$

A matrix A may itself be noninvertible, but its exponential is still invertible.

When The Exponential Of A Sum Splits

If two square matrices satisfy:

$$AB = BA,$$

then:

$$e^{(A+B)t} = e^{At}e^{Bt} = e^{Bt}e^{At}.$$

The commutativity condition is essential. It permits the binomial expansion of $(A + B)^k$ to be collected in the same way as for scalars.

A scalar multiple of the identity commutes with every matrix. This justifies:

$$e^{(\alpha I + N)t} = e^{\alpha t}e^{Nt}.$$

Where Noncommutativity First Appears

Compare the series through order t^2 .

First:

$$e^{At} = I + At + \frac{A^2 t^2}{2} + \dots$$

and:

$$e^{Bt} = I + Bt + \frac{B^2 t^2}{2} + \dots$$

Multiply while preserving order:

$$\begin{aligned} e^{At} e^{Bt} &= I + (A + B)t \\ &\quad + \left(\frac{A^2}{2} + AB + \frac{B^2}{2} \right) t^2 + \dots \end{aligned}$$

Now expand the exponential of the sum:

$$\begin{aligned} e^{(A+B)t} &= I + (A + B)t + \frac{(A + B)^2 t^2}{2} + \dots \\ &= I + (A + B)t \\ &\quad + \left(\frac{A^2}{2} + \frac{AB}{2} + \frac{BA}{2} + \frac{B^2}{2} \right) t^2 + \dots \end{aligned}$$

The quadratic coefficients agree when:

$$AB = BA.$$

If $AB \neq BA$, the two series generally differ beginning at order t^2 .

A Concrete Noncommuting Pair

Let:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Calculate AB :

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Calculate BA :

$$BA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Therefore:

$$AB \neq BA.$$

For this pair, one must not replace $e^{(A+B)t}$ by $e^{At}e^{Bt}$.

Transpose And Exponential

Transposition commutes with sums and satisfies:

$$(A^k)^\top = (A^\top)^k.$$

Transpose the defining series term by term:

$$\begin{aligned} (e^{At})^\top &= \left(I + At + \frac{A^2 t^2}{2!} + \cdots \right)^\top \\ &= I + A^\top t + \frac{(A^\top)^2 t^2}{2!} + \cdots. \end{aligned}$$

The transposed series is the defining series for $e^{A^\top t}$, so:

$$\boxed{(e^{At})^\top = e^{A^\top t}}.$$

A Reliable Computation Workflow

Step 1: Confirm that A is square and constant. This chapter's formulas assume a constant square matrix.

Step 2: Look for an immediate structure. Check whether A is diagonal, nilpotent, or of the form $\alpha I + N$.

Step 3: Calculate the characteristic polynomial. Find the eigenvalues and their algebraic multiplicities.

Step 4: Choose the reduced polynomial. For $n \times n$:

$$r_t(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1}.$$

Step 5: Build the interpolation conditions. For distinct eigenvalues, match values. For a repeated eigenvalue, also match the required derivatives with respect to λ .

Step 6: Solve for every coefficient function. Keep the exponentials such as $e^{\lambda_j t}$ visible during elimination.

Step 7: Evaluate $r_t(A)$. Replace scalar constants by multiples of I and preserve matrix powers.

Step 8: Verify the result. Check both conditions together:

$$E(0) = I, \quad E'(t) = AE(t).$$

For an IVP, also multiply by the initial vector and check its value at the specified initial time.

Common Mistake: Exponentiating Each Entry

The matrix exponential is defined by matrix powers:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \dots$$

Applying the scalar exponential to each entry produces a different operation unless special structure proves otherwise.

Common Mistake: Splitting A Sum Without Checking

The identity:

$$e^{(A+B)t} = e^{At} e^{Bt}$$

requires $AB = BA$. Check the two products explicitly before using it.

Common Mistake: Forgetting The Identity Term

The first series term is:

$$A^0 = I.$$

Therefore $e^{A0} = I$. Starting the series with At gives the wrong initial value and cannot solve the normalized matrix IVP.

Common Mistake: Stopping A Series Too Early

The series stops only when a power of the matrix is exactly zero.

If $N^2 \neq 0$ but $N^3 = 0$, then:

$$e^{Nt} = I + Nt + \frac{N^2 t^2}{2!}.$$

Dropping the quadratic term would be incorrect.

Common Mistake: Differentiating In Time During Interpolation

Repeated-eigenvalue matching differentiates:

$$r_t(\lambda)$$

with respect to λ , treating t as constant. Differentiating the coefficient functions with respect to t answers a different question.

Common Mistake: Ignoring The Initial-Time Shift

For data at $t = t_0$:

$$\mathbf{x}(t_0) = \mathbf{c},$$

the solution is:

$$\boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{c}}.$$

Using $e^{At}\mathbf{c}$ works directly only when $t_0 = 0$.

Block 7 Summary

For one fixed matrix:

$$e^{At}e^{As} = e^{A(t+s)}$$

and:

$$(e^{At})^{-1} = e^{-At}.$$

For two matrices:

$$e^{(A+B)t} = e^{At}e^{Bt}$$

is guaranteed when $AB = BA$. A finished computation should satisfy $E(0) = I$ and $E' = AE$.

Original Practice Set

Practice Problems

Problem 1: Build A Truncated Matrix Series. Let:

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}.$$

Calculate A^2 and A^3 . Then write the matrix exponential series through the term containing t^3 .

Problem 2: Exponentiate A Diagonal Matrix. Find e^{Dt} for:

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Problem 3: Exponentiate A Three-By-Three Nilpotent Matrix. Let:

$$N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Calculate the powers needed to find e^{Nt} , and then write the exponential explicitly.

Problem 4: Use A Scalar-Nilpotent Split. Find e^{At} for:

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}.$$

Problem 5: Use Distinct-Eigenvalue Interpolation. Find e^{At} for:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

by reducing the exponential to $a_0(t)I + a_1(t)A$.

Problem 6: Use A Repeated Eigenvalue. Find e^{At} for:

$$A = \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix}.$$

Show the repeated-eigenvalue value and derivative conditions.

Problem 7: Use The Even-Odd Power Pattern. Let:

$$K = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}.$$

Find e^{Kt} by calculating K^2 and separating the even and odd series terms.

Problem 8: Verify A Proposed Exponential. Verify that:

$$E(t) = e^{-2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

is e^{At} for:

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}.$$

Check both $E(0) = I$ and $E' = AE$.

Problem 9: Solve An IVP With A Nonzero Initial Time. Solve:

$$\mathbf{x}' = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(1) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Problem 10: Check The Semigroup And Inverse Laws. For:

$$N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix},$$

use the explicit formula for e^{Nt} to verify:

$$e^{Nt}e^{Ns} = e^{N(t+s)}$$

and:

$$(e^{Nt})^{-1} = e^{-Nt}.$$

Problem 11: Split A Commuting Sum. Let:

$$A = 2I, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Verify $AB = BA$, and use the commuting-sum law to find $e^{(A+B)t}$.

Problem 12: Diagnose Four Exponential Errors. Correct each statement.

1. For every matrix $A = [a_{ij}]$, the entries of e^{At} are $e^{a_{ij}t}$.
2. For all square matrices A and B , $e^{(A+B)t} = e^{At}e^{Bt}$.
3. At a repeated eigenvalue, the required extra condition is obtained by differentiating $r_t(\lambda)$ with respect to t .
4. If $\mathbf{x}(2) = \mathbf{c}$ for $\mathbf{x}' = A\mathbf{x}$, then $\mathbf{x}(t) = e^{At}\mathbf{c}$.

Worked Answers: Problems 1 To 4

Answer 1. The matrix exponential begins:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

First calculate $A^2 = AA$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + (-1)(2) & 1(-1) + (-1)(0) \\ 2(1) + 0(2) & 2(-1) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} -1 & -1 \\ 2 & -2 \end{bmatrix}. \end{aligned}$$

Now calculate $A^3 = A^2 A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} -1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -1(1) + (-1)(2) & -1(-1) + (-1)(0) \\ 2(1) + (-2)(2) & 2(-1) + (-2)(0) \end{bmatrix} \\ &= \begin{bmatrix} -3 & 1 \\ -2 & -2 \end{bmatrix}. \end{aligned}$$

Substitute the powers into the series through t^3 :

$$\begin{aligned} e^{At} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + t \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \\ &\quad + \frac{t^2}{2} \begin{bmatrix} -1 & -1 \\ 2 & -2 \end{bmatrix} + \frac{t^3}{6} \begin{bmatrix} -3 & 1 \\ -2 & -2 \end{bmatrix} + O(t^4). \end{aligned}$$

Adding I , At , $A^2t^2/2!$, and $A^3t^3/3!$ entry by entry gives:

$$e^{At} = \begin{bmatrix} 1 + t - \frac{t^2}{2} - \frac{t^3}{2} & -t - \frac{t^2}{2} + \frac{t^3}{6} \\ 2t + t^2 - \frac{t^3}{3} & 1 - t^2 - \frac{t^3}{3} \end{bmatrix} + O(t^4).$$

This is a truncated series, not yet the exact closed form.

Answer 2. For a diagonal matrix:

$$D = \text{diag}(d_1, d_2, d_3),$$

the exponential is:

$$e^{Dt} = \text{diag}(e^{d_1t}, e^{d_2t}, e^{d_3t}).$$

Here the diagonal entries are -2 , 1 , and 0 . Therefore:

$$e^{Dt} = \begin{bmatrix} e^{-2t} & 0 & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^{0t} \end{bmatrix}.$$

Since $e^{0t} = 1$:

$$e^{Dt} = \begin{bmatrix} e^{-2t} & 0 & 0 \\ 0 & e^t & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Answer 3. Start with:

$$N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Calculate $N^2 = NN$:

$$\begin{aligned} N^2 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Multiply once more:

$$\begin{aligned} N^3 &= N^2N \\ &= \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \\ &= 0. \end{aligned}$$

Since $N^3 = 0$, the exponential stops after the N^2 term:

$$e^{Nt} = I + Nt + \frac{N^2t^2}{2}.$$

Since $N^3 = 0$, write each surviving term:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$Nt = \begin{bmatrix} 0 & t & 0 \\ 0 & 0 & 2t \\ 0 & 0 & 0 \end{bmatrix},$$

and:

$$\frac{N^2 t^2}{2} = \begin{bmatrix} 0 & 0 & t^2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Add corresponding entries:

$$e^{Nt} = \begin{bmatrix} 1 & t & t^2 \\ 0 & 1 & 2t \\ 0 & 0 & 1 \end{bmatrix}.$$

Answer 4. Separate A into a scalar part and a nilpotent part:

$$A = -3I + N,$$

where:

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The nilpotent part is:

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Calculate:

$$N^2 = 0.$$

Therefore:

$$e^{Nt} = I + Nt.$$

Substitute I and N :

$$\begin{aligned} e^{Nt} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + t \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Because $-3I$ commutes with N :

$$e^{At} = e^{-3t} e^{Nt}.$$

Hence:

$$e^{At} = e^{-3t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Worked Answers: Problems 5 To 8

Answer 5. For:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

construct:

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 4 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= (\lambda - 1)(\lambda - 4) - (-2)(-2) \\ &= (\lambda - 1)(\lambda - 4) - 4. \end{aligned}$$

Expand:

$$\begin{aligned} p_A(\lambda) &= \lambda^2 - 5\lambda + 4 - 4 \\ &= \lambda^2 - 5\lambda \\ &= \lambda(\lambda - 5). \end{aligned}$$

The eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 5.$$

Use:

$$r_t(\lambda) = a_0 + a_1\lambda.$$

For the eigenvalues 0 and 5, first impose the condition at $\lambda = 0$:

$$r_t(0) = e^{0t} = 1.$$

Substitute $\lambda = 0$:

$$a_0 = 1.$$

At $\lambda = 5$:

$$r_t(5) = e^{5t}.$$

Substitute $\lambda = 5$:

$$a_0 + 5a_1 = e^{5t}.$$

Use $a_0 = 1$:

$$1 + 5a_1 = e^{5t}.$$

Subtract 1:

$$5a_1 = e^{5t} - 1.$$

Divide by 5:

$$a_1 = \frac{e^{5t} - 1}{5}.$$

The reduced polynomial evaluated at A is:

$$e^{At} = I + \frac{e^{5t} - 1}{5}A.$$

Insert I and A :

$$\begin{aligned} e^{At} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{e^{5t} - 1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 4 + e^{5t} & 2e^{5t} - 2 \\ 2e^{5t} - 2 & 1 + 4e^{5t} \end{bmatrix}. \end{aligned}$$

Hence:

$$e^{At} = \frac{1}{5} \begin{bmatrix} 4 + e^{5t} & 2e^{5t} - 2 \\ 2e^{5t} - 2 & 1 + 4e^{5t} \end{bmatrix}.$$

Answer 6. Construct the characteristic polynomial for:

$$A = \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix}.$$

First:

$$\lambda I - A = \begin{bmatrix} \lambda & -1 \\ 4 & \lambda - 4 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= \lambda(\lambda - 4) - (-1)(4) \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2. \end{aligned}$$

Thus $\lambda = 2$ has multiplicity 2.

For this repeated eigenvalue, use:

$$r_t(\lambda) = a_0 + a_1\lambda.$$

The value condition is:

$$r_t(2) = e^{2t}.$$

Substitute $\lambda = 2$:

$$a_0 + 2a_1 = e^{2t}.$$

For the derivative condition, differentiate with respect to λ :

$$\frac{\partial r_t}{\partial \lambda} = a_1$$

and:

$$\frac{\partial}{\partial \lambda} e^{\lambda t} = te^{\lambda t}.$$

Evaluate at $\lambda = 2$:

$$a_1 = te^{2t}.$$

Substitute this into the value equation:

$$a_0 + 2te^{2t} = e^{2t}.$$

The remaining coefficient is:

$$a_0 = e^{2t}(1 - 2t).$$

Now evaluate the polynomial at A :

$$\begin{aligned} e^{At} &= a_0 I + a_1 A \\ &= e^{2t}(1 - 2t)I + te^{2t}A \\ &= e^{2t}[(1 - 2t)I + tA]. \end{aligned}$$

Substitute the matrices:

$$\begin{aligned} (1 - 2t)I + tA &= \begin{bmatrix} 1 - 2t & 0 \\ 0 & 1 - 2t \end{bmatrix} + \begin{bmatrix} 0 & t \\ -4t & 4t \end{bmatrix} \\ &= \begin{bmatrix} 1 - 2t & t \\ -4t & 1 + 2t \end{bmatrix}. \end{aligned}$$

Thus:

$$e^{At} = e^{2t} \begin{bmatrix} 1 - 2t & t \\ -4t & 1 + 2t \end{bmatrix}.$$

Answer 7. Calculate K^2 :

$$\begin{aligned} K^2 &= \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -9 & 0 \\ 0 & -9 \end{bmatrix} \\ &= -9I. \end{aligned}$$

From $K^2 = -9I$, the resulting even-odd power pattern is:

$$K^{2m} = (-9)^m I$$

and:

$$K^{2m+1} = (-9)^m K.$$

The even terms in the exponential series give:

$$\begin{aligned} \sum_{m=0}^{\infty} \frac{K^{2m} t^{2m}}{(2m)!} &= I \sum_{m=0}^{\infty} \frac{(-1)^m (3t)^{2m}}{(2m)!} \\ &= I \cos(3t). \end{aligned}$$

The odd terms give:

$$\begin{aligned} \sum_{m=0}^{\infty} \frac{K^{2m+1} t^{2m+1}}{(2m+1)!} &= \frac{K}{3} \sum_{m=0}^{\infty} \frac{(-1)^m (3t)^{2m+1}}{(2m+1)!} \\ &= \frac{K}{3} \sin(3t). \end{aligned}$$

Therefore:

$$e^{Kt} = I \cos(3t) + \frac{K}{3} \sin(3t).$$

Since:

$$\frac{K}{3} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

Combining the even and odd sums, the exponential is:

$$e^{Kt} = \begin{bmatrix} \cos(3t) & -\sin(3t) \\ \sin(3t) & \cos(3t) \end{bmatrix}.$$

Answer 8. The proposed matrix is:

$$E(t) = e^{-2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

First set $t = 0$:

$$\begin{aligned} E(0) &= e^0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I. \end{aligned}$$

Now differentiate. Let:

$$M(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Then:

$$M'(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Use the product rule:

$$\begin{aligned} E'(t) &= -2e^{-2t}M(t) + e^{-2t}M'(t) \\ &= e^{-2t} [-2M(t) + M'(t)]. \end{aligned}$$

For the proposed $E(t) = e^{-2t}M(t)$, calculate the product-rule bracket:

$$\begin{aligned} -2M + M' &= -2 \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 1 - 2t \\ 0 & -2 \end{bmatrix}. \end{aligned}$$

Thus:

$$E'(t) = e^{-2t} \begin{bmatrix} -2 & 1 - 2t \\ 0 & -2 \end{bmatrix}.$$

Now calculate $AE(t)$:

$$\begin{aligned} AE(t) &= \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} e^{-2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \\ &= e^{-2t} \begin{bmatrix} -2 & 1 - 2t \\ 0 & -2 \end{bmatrix}. \end{aligned}$$

Therefore:

$$E'(t) = AE(t).$$

Together, $E(0) = I$ and $E' = AE$ verify:

$$\boxed{E(t) = e^{At}}.$$

Worked Answers: Problems 9 To 12

Answer 9. For:

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix},$$

write:

$$A = -3I + N, \quad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Since $N^2 = 0$:

$$e^{N\tau} = I + N\tau = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix}.$$

Therefore, for any time interval τ :

$$e^{A\tau} = e^{-3\tau} \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix}.$$

The initial time is $t_0 = 1$, so use:

$$\tau = t - 1.$$

The IVP solution is:

$$\mathbf{x}(t) = e^{A(t-1)} \mathbf{x}(1).$$

Substitute the exponential and initial vector:

$$\begin{aligned}
 \mathbf{x}(t) &= e^{-3(t-1)} \begin{bmatrix} 1 & t-1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\
 &= e^{-3(t-1)} \begin{bmatrix} 2 - (t-1) \\ -1 \end{bmatrix} \\
 &= e^{-3(t-1)} \begin{bmatrix} 3-t \\ -1 \end{bmatrix}.
 \end{aligned}$$

Thus:

$$\mathbf{x}(t) = e^{-3(t-1)} \begin{bmatrix} 3-t \\ -1 \end{bmatrix}.$$

Check the initial time:

$$\begin{aligned}
 \mathbf{x}(1) &= e^0 \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\
 &= \begin{bmatrix} 2 \\ -1 \end{bmatrix}.
 \end{aligned}$$

Answer 10. For:

$$N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix},$$

we have $N^2 = 0$. Therefore:

$$e^{Nt} = I + Nt = \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

The same nilpotent formula evaluated at time s gives:

$$e^{Ns} = \begin{bmatrix} 1 & 2s \\ 0 & 1 \end{bmatrix}.$$

Multiply:

$$\begin{aligned} e^{Nt}e^{Ns} &= \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2s \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2s + 2t \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2(t+s) \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

But:

$$e^{N(t+s)} = \begin{bmatrix} 1 & 2(t+s) \\ 0 & 1 \end{bmatrix}.$$

Hence:

$$\boxed{e^{Nt}e^{Ns} = e^{N(t+s)}}.$$

For the inverse, replace t by $-t$:

$$e^{-Nt} = \begin{bmatrix} 1 & -2t \\ 0 & 1 \end{bmatrix}.$$

To verify the inverse relation, multiply:

$$\begin{aligned}e^{Nt}e^{-Nt} &= \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2t \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I.\end{aligned}$$

Therefore:

$$\boxed{(e^{Nt})^{-1} = e^{-Nt}}.$$

Answer 11. The matrices are:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Calculate AB :

$$\begin{aligned}AB &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.\end{aligned}$$

Calculate BA :

$$\begin{aligned}BA &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.\end{aligned}$$

The two calculated products are equal:

$$AB = BA.$$

The commuting-sum law applies:

$$e^{(A+B)t} = e^{At}e^{Bt}.$$

Since $A = 2I$:

$$e^{At} = e^{2t}I.$$

Also $B^2 = 0$, so:

$$e^{Bt} = I + Bt = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Multiply:

$$\begin{aligned} e^{(A+B)t} &= e^{2t}I \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \\ &= e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Therefore:

$$e^{(A+B)t} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Answer 12.

Statement 1. The matrix exponential is not generally calculated entry by entry. Its definition is:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \dots$$

The powers are matrix products. Entrywise exponentiation is valid only when a special structure, such as a diagonal matrix, justifies it.

Statement 2. The splitting law requires commutativity.

The corrected statement is:

$$AB = BA \implies e^{(A+B)t} = e^{At} e^{Bt}.$$

Without $AB = BA$, the products AB and BA appear differently in the quadratic series terms.

Statement 3. At a repeated eigenvalue, the interpolation variable is λ . Hold t fixed and differentiate with respect to λ :

$$\frac{\partial^q r_t}{\partial \lambda^q}(\lambda_0) = \frac{\partial^q}{\partial \lambda^q} e^{\lambda t} \Big|_{\lambda=\lambda_0}.$$

Differentiating with respect to t does not provide the required matching condition.

Statement 4. The initial time is $t_0 = 2$. The time interval in the matrix exponential must therefore be $t - 2$.

The corrected solution is:

$$\mathbf{x}(t) = e^{A(t-2)} \mathbf{c}.$$

Indeed, setting $t = 2$ gives:

$$\mathbf{x}(2) = e^{A0} \mathbf{c} = I \mathbf{c} = \mathbf{c}.$$

Chapter 16 Final Summary

For a constant square matrix A :

$$e^{At} = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}.$$

It satisfies:

$$e^{A0} = I, \quad \frac{d}{dt} e^{At} = A e^{At} = e^{At} A.$$

Therefore the constant-coefficient IVP:

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{c}$$

has the solution:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{c}.$$

Easy structures should be checked first:

- diagonal matrices exponentiate along the diagonal
- if $N^r = 0$, the series for e^{Nt} stops after N^{r-1}
- if $A = \alpha I + N$, then $e^{At} = e^{\alpha t} e^{Nt}$

Cayley-Hamilton reduces the exponential to:

$$e^{At} = a_0(t)I + a_1(t)A + \cdots + a_{n-1}(t)A^{n-1}.$$

For distinct eigenvalues, match:

$$r_t(\lambda_j) = e^{\lambda_j t}.$$

For an eigenvalue of multiplicity m , also match derivatives with respect to λ through order $m - 1$.

Complex eigenvalues can produce real sine-cosine matrices. If $K^2 = -\beta^2 I$, then:

$$e^{Kt} = I \cos(\beta t) + \frac{K}{\beta} \sin(\beta t).$$

The core exponential laws are:

$$e^{At}e^{As} = e^{A(t+s)}, \quad (e^{At})^{-1} = e^{-At}.$$

For two different matrices, the split:

$$e^{(A+B)t} = e^{At}e^{Bt}$$

is guaranteed when $AB = BA$.

The chapter's central lesson is:

treat e^{At} as a matrix function built from matrix powers: exploit structure first, preserve factor order, match repeated eigenvalues in λ , and verify every final formula with $E(0) = I$ and $E' = AE$.